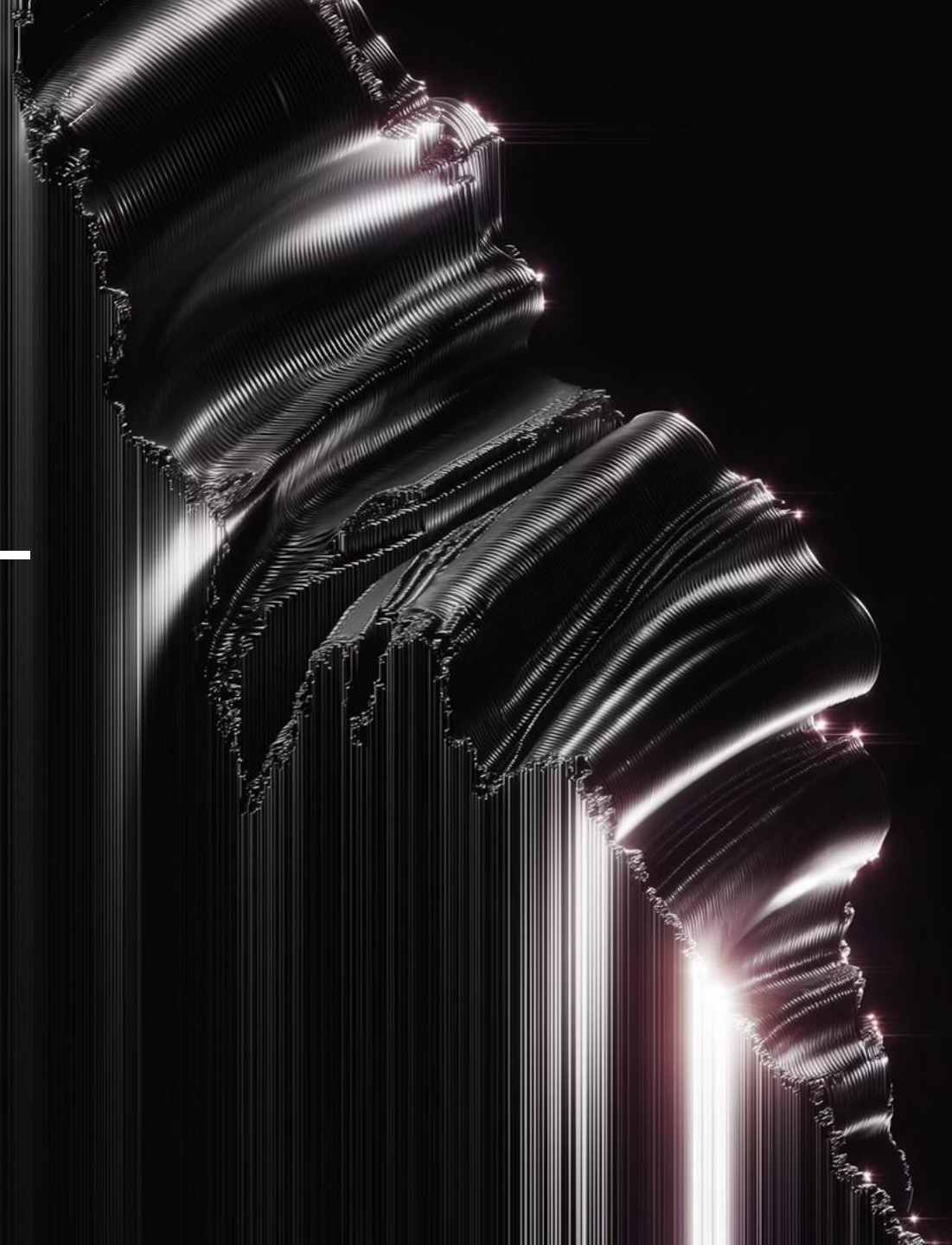


Payload & tool protection and Anti- Analysis techniques for red teaming



Whoami

Antonio D. Seri – Andi

Senior Red Team Operator

Studies & Certs:

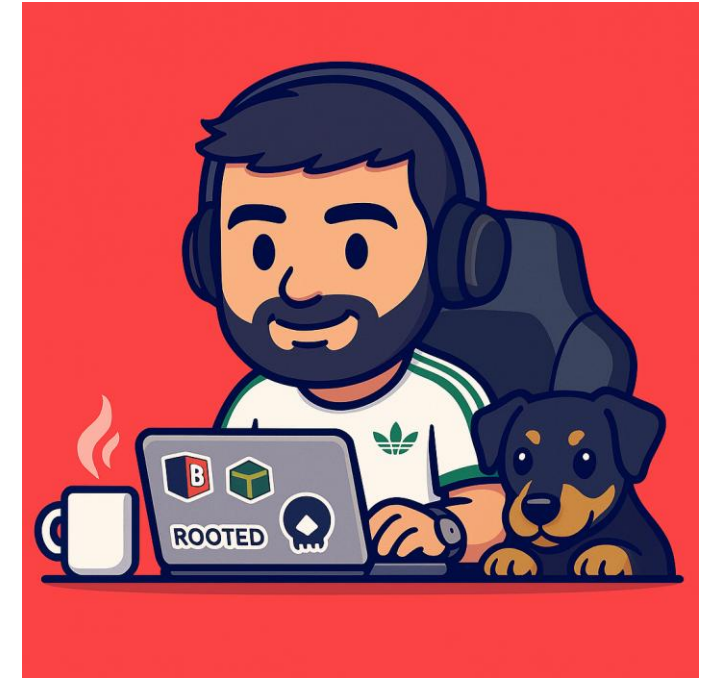
- CRTL
- CRTO
- Malware Development MalDev Academy
- Phishing Development MalDev Academy
- Master's Degree in Cybersecurity Management and Ethical Hacking
- ODPC de WKL [In Progress]

Contact & links:

cOnfig17.com

[Linkedin.com/in/antoniodseri47](https://linkedin.com/in/antoniodseri47)

github.com/cOnfig-17

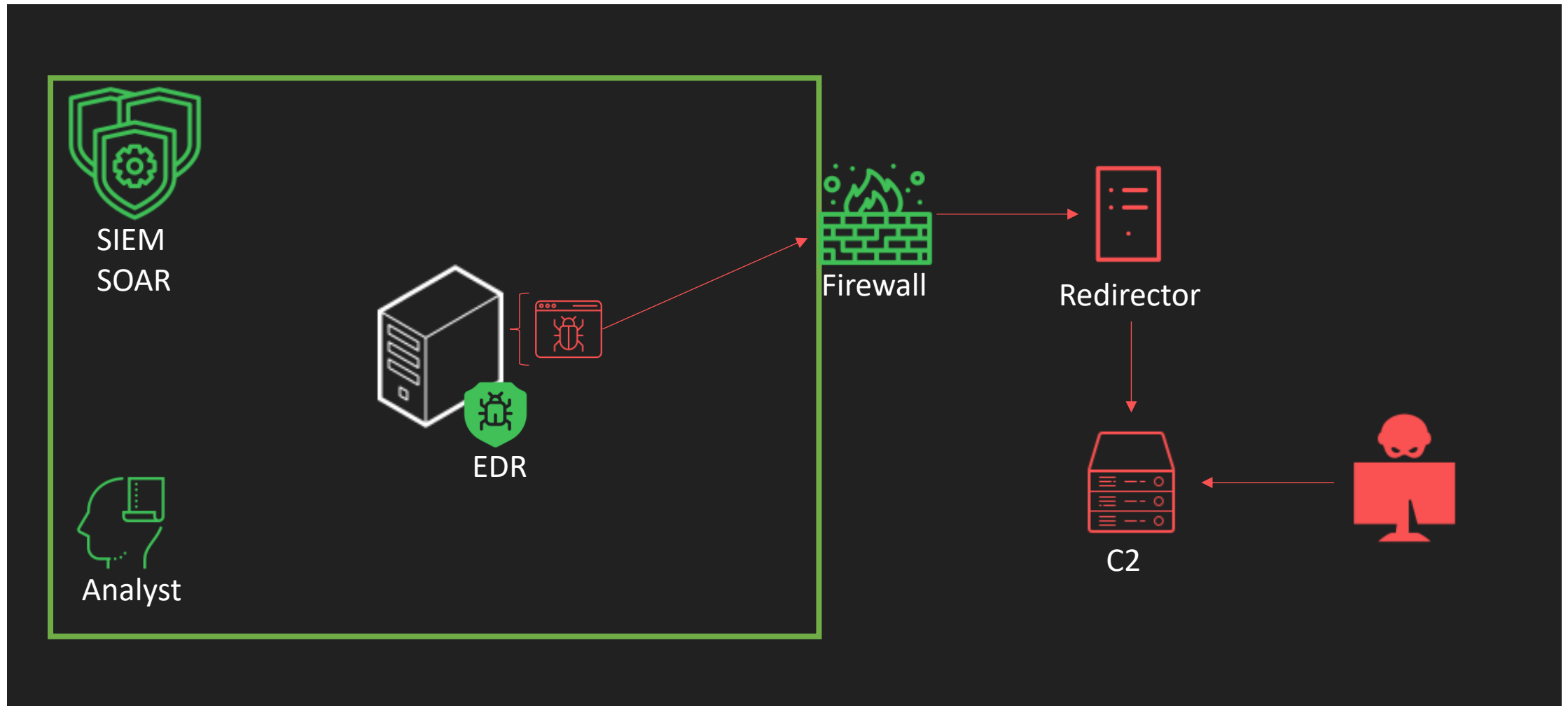


stratesys

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1. Introduction to the problem(S)



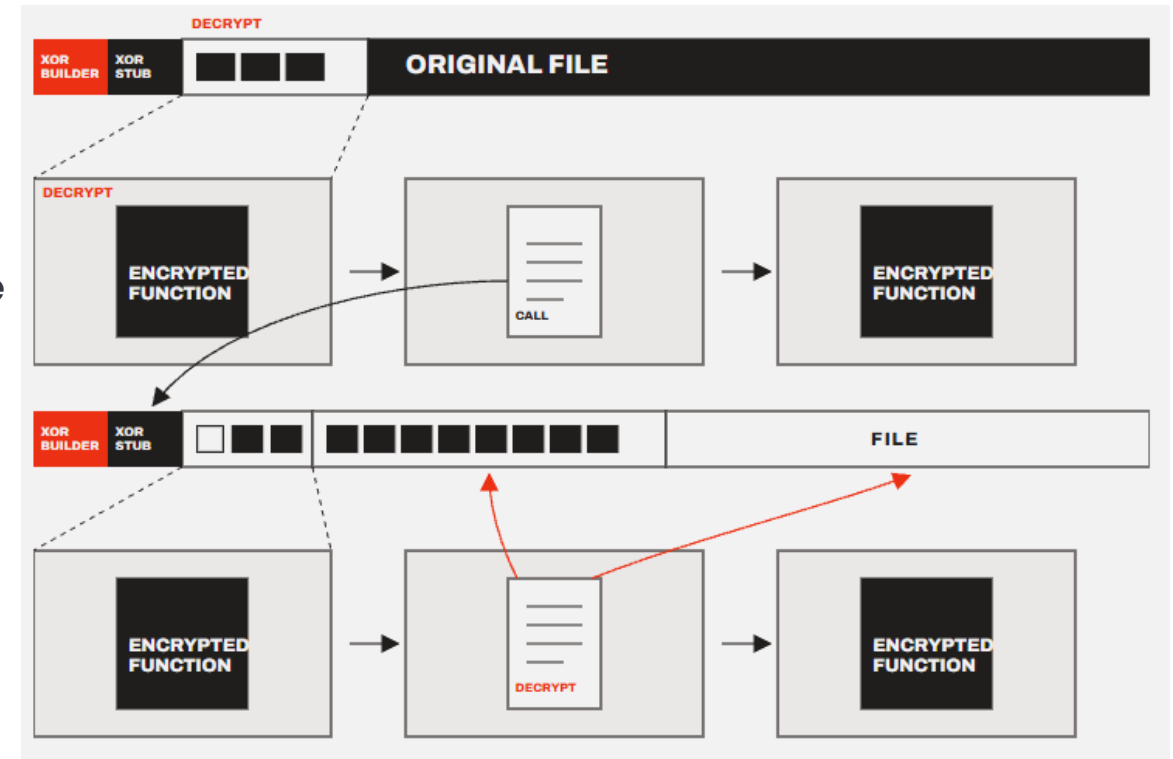
1. Introduction to the problem(S)

Problems to solve:

- The use of AI allows flawed analysis procedures to pass as robust ones. -> Make limitations visible
- Limitations in the use of certain tools and the temporary inability to develop new ones due to time constraints.

Goals and limitations to solve:

- Easy implementation
- Opsec methodology
- Preventive solutions



2. Counterintelligence & Anti-Analysis

- Stay one step ahead in the analysis
- Limit the target's ability to act.
- First analysis and block visibility on N1.
- Block LLM-AI analysis.
- Prevent bot analysis capabilities.
- Not just-OPSEC.

COMMANDER IN CHIEF PACIFIC



CINCPAC PURPLE DRAGON report coversheet (artwork by USN)

~~NOT RELEASABLE TO FOREIGN NATIONALS~~

35

(b) (3) - P.L. 86-36

~~TOP SECRET UMBRA~~

2. Counterintelligence & Anti-Analysis

Some tips, tricks and recommendations:

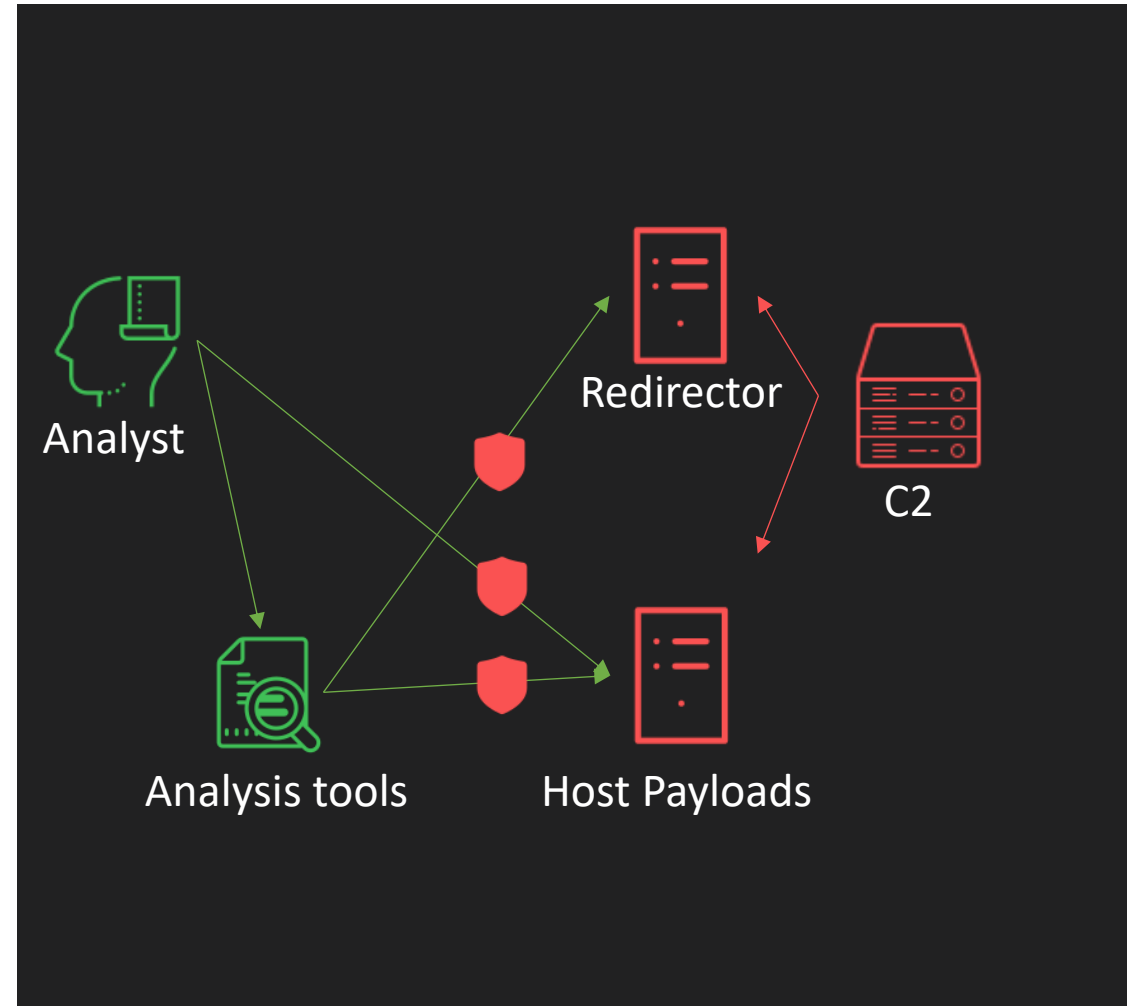
- UserAgent & AntiCrawling
 - Anomalous User Agents
 - Discontinued Software User Agents
 - Signed User Agents
 - Bots & Crawling

```
github.com/arcjet/well-known-bots/blob/main/well-known-bots.json

well-known-bots / well-known-bots.json

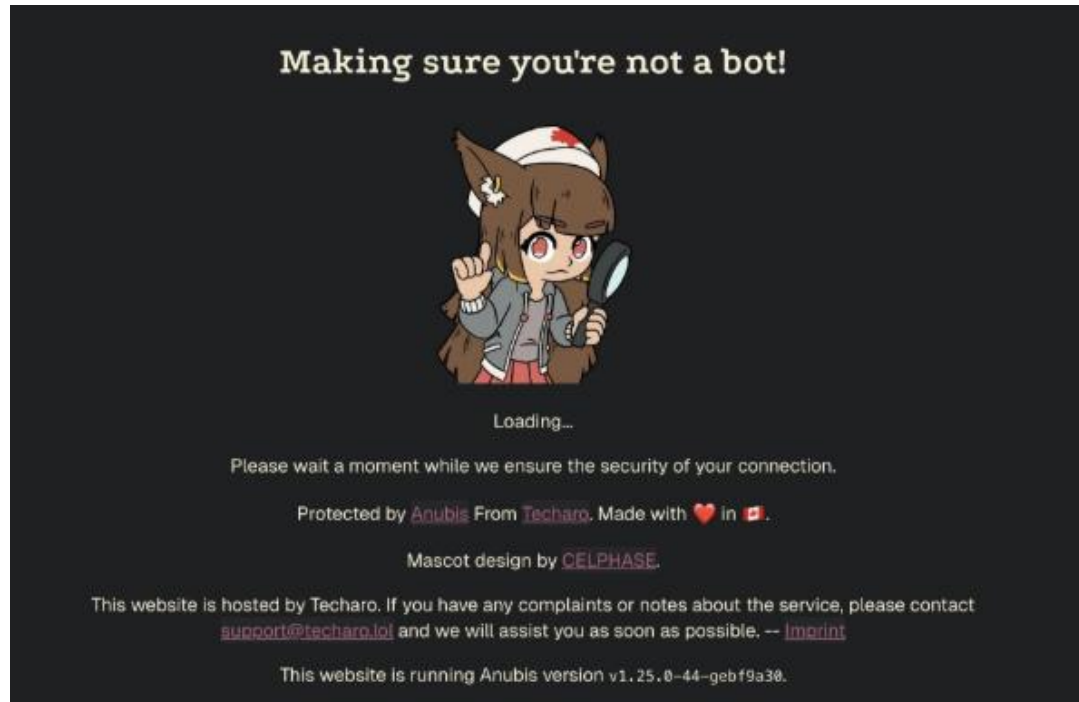
18636 lines (18636 loc) · 427 KB

"Mozilla/5.0 (compatible; Googlebot/2.1; +http://www.google.com/bot.html)",
"Mozilla/5.0 (iPhone; CPU iPhone OS 6_0 like Mac OS X) AppleWebKit/536.26 (KHTML, like Gecko) Version/6.0 Mobile",
"Mozilla/5.0 (iPhone; CPU iPhone OS 8_3 like Mac OS X) AppleWebKit/537.36 (KHTML, like Gecko) Version/8.0 Mobile",
"Mozilla/5.0 (iPhone; CPU iPhone OS 8_3 like Mac OS X) AppleWebKit/600.1.4 (KHTML, like Gecko) Version/8.0 Mobile",
"Mozilla/5.0 (Linux; Android 6.0.1; Nexus 5X Build/MMB29P) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/27.0.14",
"Mozilla/5.0 (Linux; Android 6.0.1; Nexus 5X Build/MMB29P) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/41.0.22",
"Mozilla/5.0 AppleWebKit/537.36 (KHTML, like Gecko; compatible; Googlebot/2.1; +http://www.google.com/bot.html)"
```



2. Counterintelligence & Anti-Analysis

Antibot infra

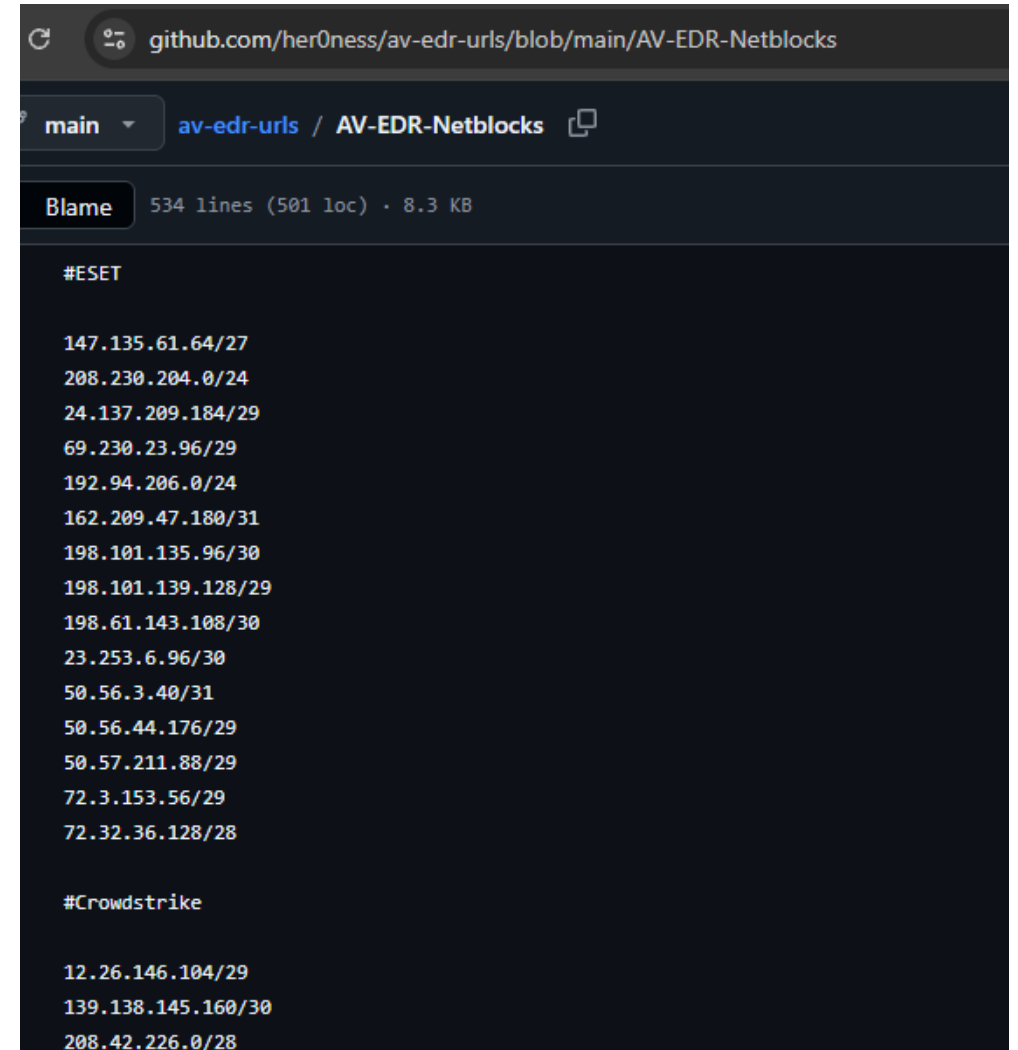


<https://github.com/TecharoHQ/anubis>

2. Counterintelligence & Anti-Analysis

IP Filtering

- Public repositories
<https://github.com/herOness/av-edr-urls>
- Your Own Methodology for Discovery
 - Targeting Analysis solutions
 - Honeypotting → C2 & Evilginx
 - Search Netblocks



The screenshot shows a GitHub repository page for 'av-edr-urls' by user 'herOness'. The file 'AV-EDR-Netblocks' is selected, showing 534 lines of code. The content is a list of IP ranges categorized by vendor.

```
#ESET

147.135.61.64/27
208.230.204.0/24
24.137.209.184/29
69.230.23.96/29
192.94.206.0/24
162.209.47.180/31
198.101.135.96/30
198.101.139.128/29
198.61.143.108/30
23.253.6.96/30
50.56.3.40/31
50.56.44.176/29
50.57.211.88/29
72.3.153.56/29
72.32.36.128/28

#CrowdStrike

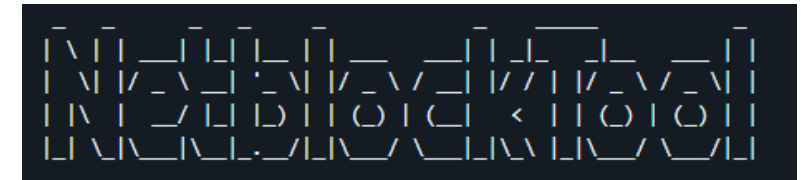
12.26.146.104/29
139.138.145.160/30
208.42.226.0/28
```

2. Counterintelligence & Anti-Analysis

Check NetBlocks of analysis providers via ARIN

- American Registry for Internet Numbers

```
(root@kali)-[/home/kali/Desktop]
# python3 arin.py -v -4 "kaspersky"
[*] Buscando entidades ARIN para: kaspersky*
[*] 6 entidades. Recuperando redes...
[1/6] KASPERSKY LAB                redes: 1
[2/6] Kaspersky Lab B.V.           redes: 3
[3/6] KASPERSKY LAB INC.           redes: 0
[4/6] Kaspersky Lab Switzerland GmbH redes: 1
[5/6] Kaspersky Lab Switzerland GmbH redes: 1
[6/6] Kaspersky Lab, Inc           redes: 1
[+] 66.110.49.0/25                HIGH KASPERSKY-TATAC
[+] 66.110.49.128/25              HIGH KASPERSKY-TATAC
[+] 74.118.24.176/28              HIGH LIGHTOWER-KASPERSKY-74-118-24-17
[+] 66.46.92.160/27               MEDIUM NET-66-46-92-160-27
[+] 71.162.124.160/29             MEDIUM FTTP
[+] 216.13.43.172/30              MEDIUM NET-216-13-43-172-30
[*] 6 netblocks unicos
66.46.92.160/27
66.110.49.0/25
66.110.49.128/25
71.162.124.160/29
74.118.24.176/28
216.13.43.172/30
```



 NetSPI™

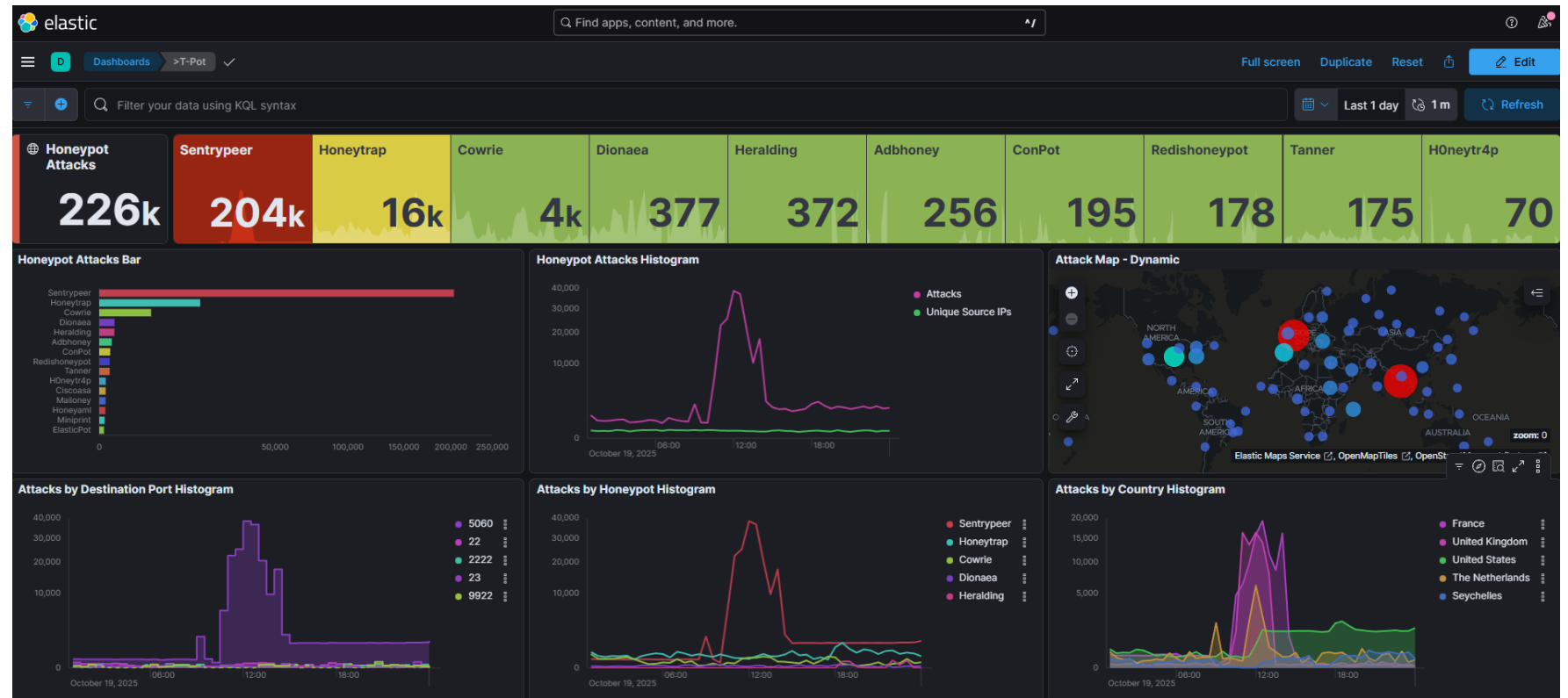


2. Counterintelligence & Anti-Analysis

T – Pot “The all in one multi honeypot platform”

→ Multiple Honeypots

→ Easy deployment



→ <https://github.com/telekom-security/tpotce>

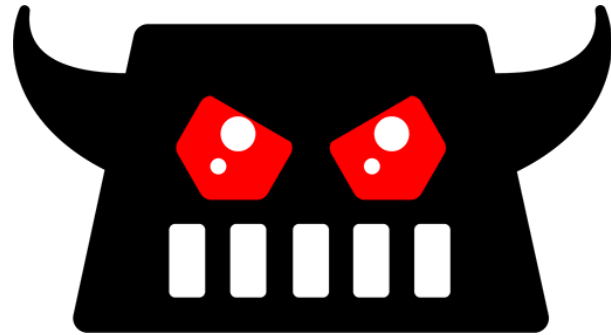
2. Counterintelligence & Anti-Analysis

Honeypotting Evilginx

→ Abussing Certificate Transparency

→ Let's Encrypt

→ Use Any C2 clearly and Intercept traffic



```
[21:31:49] [war] blacklist: request from ip address '147.182.149.75' was blocked
[21:31:49] [war] blacklist: request from ip address '64.23.218.208' was blocked
[21:31:49] [war] blacklist: request from ip address '46.101.111.185' was blocked
[21:31:49] [war] blacklist: request from ip address '165.227.39.235' was blocked
[21:31:49] [war] blacklist: request from ip address '147.182.149.75' was blocked
[21:31:49] [war] blacklist: request from ip address '138.197.191.87' was blocked
[21:31:50] [war] blacklist: request from ip address '104.244.74.39' was blocked
[21:31:50] [war] blacklist: request from ip address '104.244.74.39' was blocked
[21:31:50] [war] blacklist: request from ip address '46.101.111.185' was blocked
[21:31:50] [war] blacklist: request from ip address '104.244.74.39' was blocked
[21:31:50] [war] blacklist: request from ip address '64.23.218.208' was blocked
[21:31:50] [war] blacklist: request from ip address '147.182.149.75' was blocked
[21:31:50] [war] blacklist: request from ip address '104.244.74.39' was blocked
[21:31:50] [war] blacklist: request from ip address '104.244.74.39' was blocked
[21:31:50] [war] blacklist: request from ip address '165.227.39.235' was blocked
[21:31:50] [war] blacklist: request from ip address '159.89.174.87' was blocked
[21:31:50] [war] blacklist: request from ip address '104.244.74.39' was blocked
[21:31:50] [war] blacklist: request from ip address '104.244.74.39' was blocked
[21:31:50] [war] blacklist: request from ip address '104.244.74.39' was blocked
[21:31:50] [war] blacklist: request from ip address '138.197.191.87' was blocked
[21:31:51] [war] blacklist: request from ip address '147.182.149.75' was blocked
[21:31:51] [war] blacklist: request from ip address '46.101.111.185' was blocked
[21:31:51] [war] blacklist: request from ip address '96.41.38.202' was blocked
[21:31:51] [war] blacklist: request from ip address '64.23.218.208' was blocked
[21:31:51] [war] blacklist: request from ip address '165.227.39.235' was blocked
[21:31:51] [war] blacklist: request from ip address '138.197.191.87' was blocked
[21:31:52] [war] blacklist: request from ip address '147.182.149.75' was blocked
[21:31:52] [war] blacklist: request from ip address '46.101.111.185' was blocked
[21:31:52] [war] blacklist: request from ip address '64.23.218.208' was blocked
[21:31:52] [war] blacklist: request from ip address '165.227.39.235' was blocked
[21:31:52] [war] blacklist: request from ip address '138.197.191.87' was blocked
[21:31:52] [war] blacklist: request from ip address '46.101.111.185' was blocked
[21:31:53] [war] blacklist: request from ip address '147.182.149.75' was blocked
[21:31:53] [war] blacklist: request from ip address '64.23.218.208' was blocked
[21:31:53] [war] blacklist: request from ip address '165.227.39.235' was blocked
[21:31:53] [war] [dropbox] unauthorized request: https://fp. .xyz/ (Mozilla/5.0 (Windows NT 10.0; Win64; x
4) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36) [172.233.62.72]
[21:31:53] [war] blacklisted ip address: 172.233.62.72
```

2. Counterintelligence & Anti-Analysis



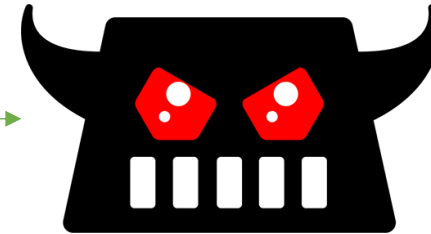
```
[10:45:27] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.0.0 Safari/537.36 AppEngine-Google; (+http://code.google.com/appengine; appid: s~virustotalcloud)) [34.127.136.67]
[10:45:27] [war] blacklisted ip address: 34.127.136.67
[10:45:27] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/114.0.0.0 Safari/537.36 Edg/114.0.1823.51) [3.144.93.15]
[10:45:27] [war] blacklisted ip address: 3.144.93.15
[10:45:27] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/114.0.0.0 Safari/537.36 Edg/114.0.1823.51) [3.150.127.118]
[10:45:27] [war] blacklisted ip address: 3.150.127.118
[10:45:28] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.0.0 Safari/537.36 AppEngine-Google; (+http://code.google.com/appengine; appid: s~virustotalcloud)) [34.127.136.68]
[10:45:28] [war] blacklisted ip address: 34.127.136.68
[10:45:28] [war] [new] unauthorized request: https://www.new.c0nfig17.com/favicon.ico (python-requests/2.33.0) [34.96.44.57]
[10:45:28] [war] blacklisted ip address: 34.96.44.57
[10:45:28] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 (Linux; Android 10; K) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/124.0.0.0 Mobile Safari/537.36) [136.70.197.14]
[10:45:28] [war] blacklisted ip address: 136.70.197.14
[10:45:33] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36 Edg/151.0.0.0) [170.64.241.17]
[10:45:33] [war] blacklisted ip address: 170.64.241.17
[10:45:37] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.0.0 Safari/537.36 AppEngine-Google; (+http://code.google.com/appengine; appid: s~virustotalcloud)) [34.127.136.66]
[10:45:37] [war] blacklisted ip address: 34.127.136.66
[10:45:37] [war] blacklist: request from ip address '34.127.136.68' was blocked
```



```
[10:29:53] [war] [new] unauthorized request: https://www.new.c0nfig17.com/phpinfo.php (Python/3.12 aiohttp/3.9.3) [4.230.144.132]
[10:29:53] [war] blacklisted ip address: 4.230.144.132
[10:29:53] [war] blacklist: request from ip address '4.230.144.132' was blocked
[10:29:53] [war] blacklist: request from ip address '4.230.144.132' was blocked
```

```
[10:33:27] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36 GeoPeeker/1.0 (+https://geopeeker.com/; support@geopeeker.com)) [52.53.149.227]
[10:33:27] [war] blacklisted ip address: 52.53.149.227
```

2. Counterintelligence & Anti-Analysis



Eres un analista de malware y te han pasado esta url [www.new.c0nfig17.com]
(<https://www.new.c0nfig17.com>) determina si ese subdominio exacto está alojando algo malicioso

Investigating whether a subdomain hosts malicious content.

Voy a investigar la reputación de ese subdominio.

Web buscada, leída una página ▾

```
Web busca [10:17:37] [war] [new] unauthorized request: https://www.new.c0nfig17.com/robots.txt (Mozilla/5.0 AppleWebKit/537.36 (KHTML, like Gecko; compatible; Claude-User/1.0; +claude-user@anthropic.com)) [34.162.191.81]
Web busca [10:17:37] [war] blacklisted ip address: 34.162.191.81
Fetches: h [10:17:38] [war] blacklist: request from ip address '34.162.191.81' was blocked
```

Web buscada "new.c0nfig17" OR "c0nfig17" phishing evilginx >

Redactando un resumen honesto sobre el dominio sospechoso.

Veredicto: no puedo confirmar que www.new.c0nfig17.com aloje malware. Con lo que he podido ver, lo clasificaría como **sospechoso/indeterminado**, probablemente infraestructura de pruebas de red team y no una campaña criminal.

2. Counterintelligence & Anti-Analysis



```
[10:35:07] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Google-Lens) [74.125.211.6]
[10:35:07] [war] blacklisted ip address: 74.125.211.6
[10:35:07] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Google-Lens) [74.125.211.5]
[10:35:07] [war] blacklisted ip address: 74.125.211.5
```



```
[10:38:57] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 AppleWebKit/537.36 (KHTML, like Gecko); compatible; ChatGPT-User/1.0; +https://openai.com/bot) [9.129.58.31]
[10:38:57] [war] blacklisted ip address: 9.129.58.31
[10:39:16] [war] [new] unauthorized request: https://www.new.c0nfig17.com/robots.txt (Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/131.0.0.0 Safari/537.36; compatible; OAI-SearchBot/1.0; +https://openai.com/searchbot) [51.8.102.224]
[10:39:16] [war] blacklisted ip address: 51.8.102.224
```



```
[10:44:01] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/133.0.0.0 Safari/537.36 Edg/133.0.0.0) [91.230.225.18]
[10:44:01] [war] blacklisted ip address: 91.230.225.18
```

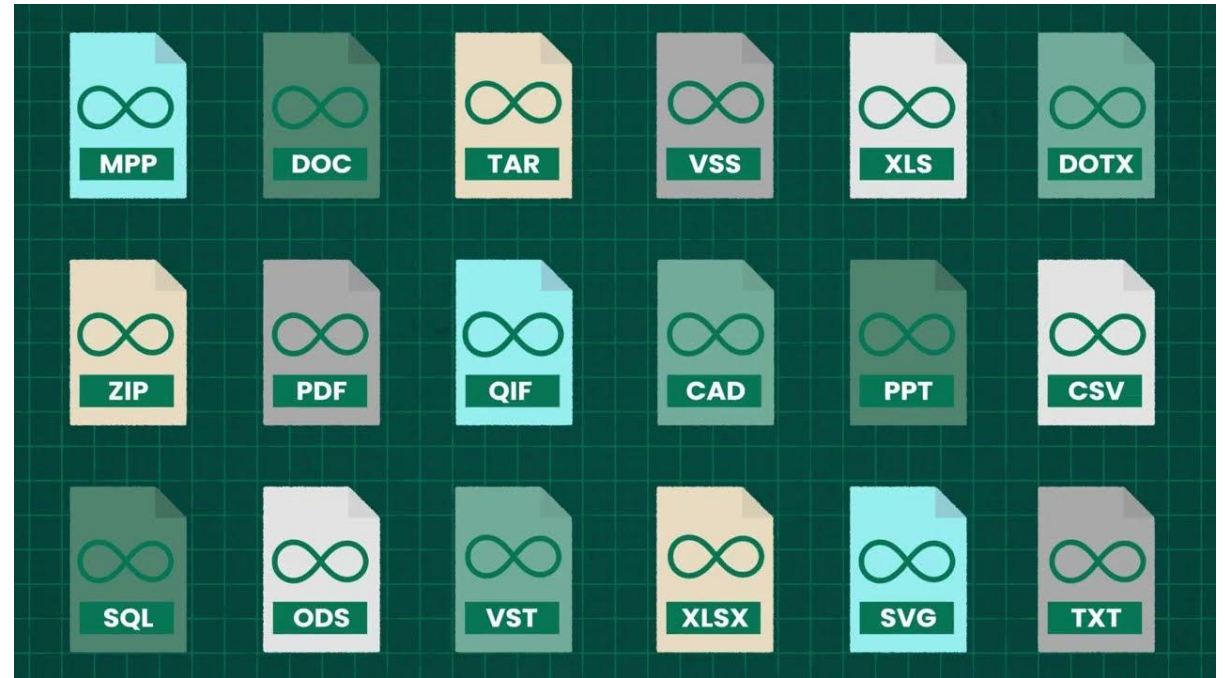


```
[10:54:11] [war] blacklisted ip address: 186.109.41.117
[10:54:11] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36) [179.238.47.30]
[10:54:11] [war] blacklisted ip address: 179.238.47.30
[10:54:15] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36) [92.71.67.194]
[10:54:15] [war] blacklisted ip address: 92.71.67.194
[10:54:20] [war] [new] unauthorized request: https://www.new.c0nfig17.com/ (curl/7.88.1) [34.186.82.14]
[10:54:20] [war] blacklisted ip address: 34.186.82.14
[10:54:36] [war] blacklist: request from ip address '34.186.82.14' was blocked
```

3. Payload protection

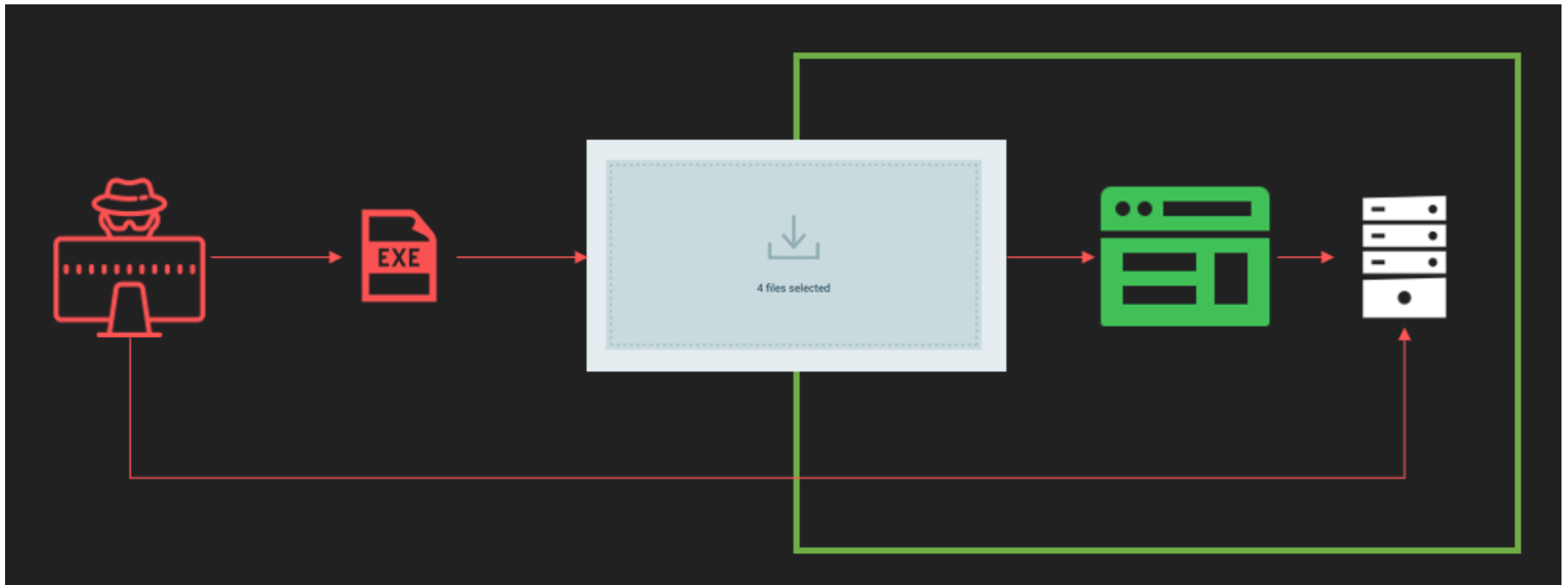
1. CDR evasion via File Bloating for Initial Access

2. IPv4 Obfuscation and IT environments



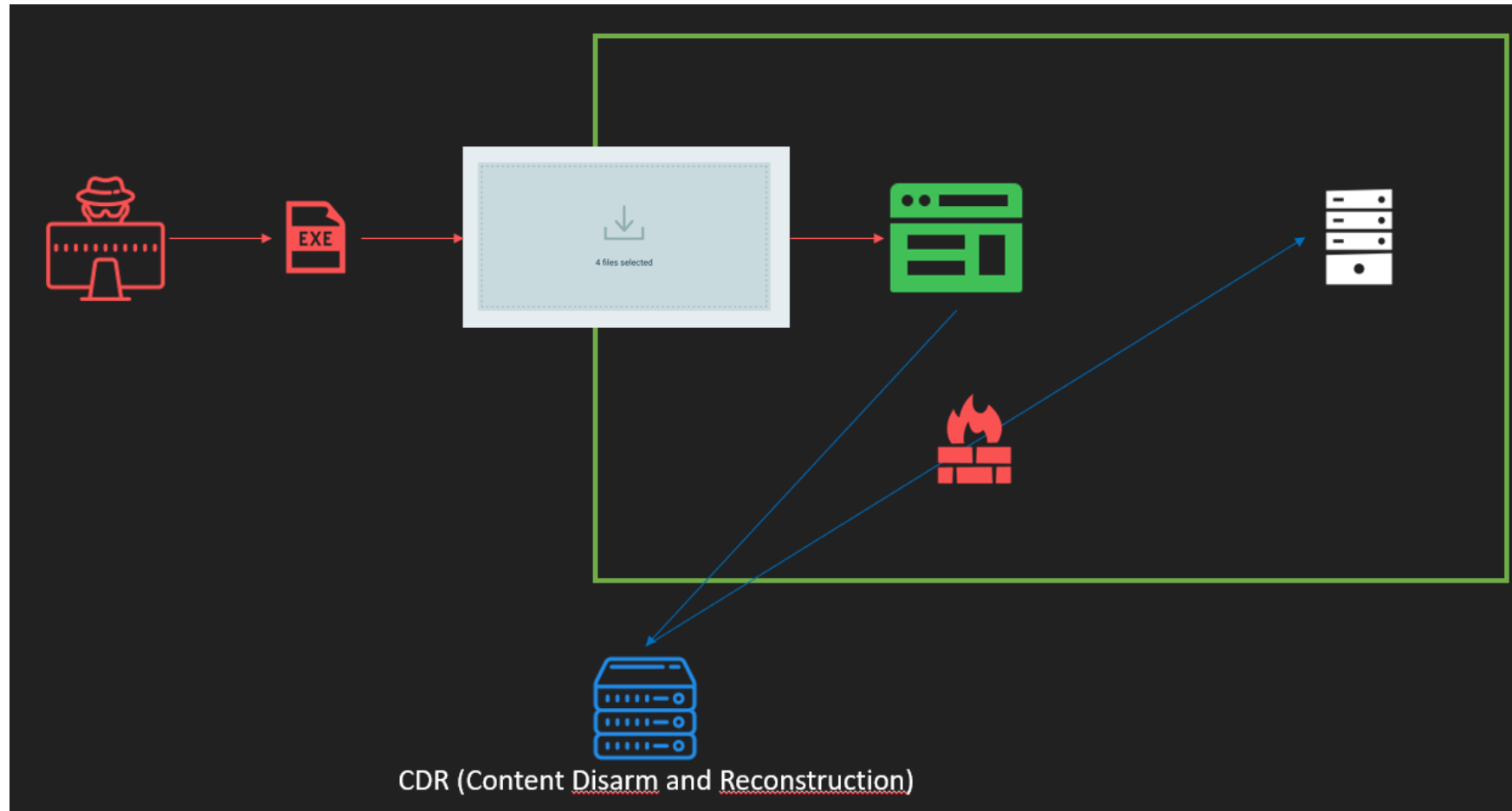
3. Payload protection

1. CDR evasión via File Bloating for Initial Access



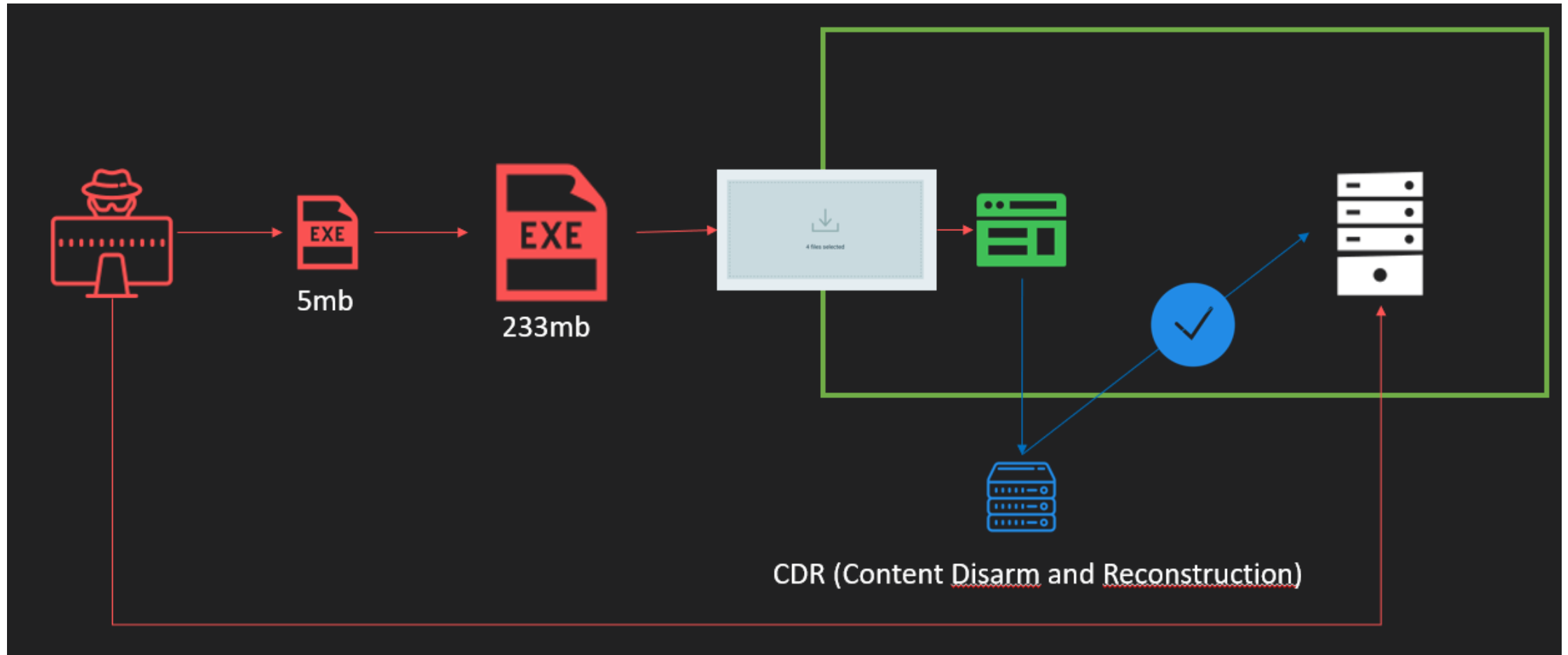
3. Payload protection

1. CDR evasión via File Bloating for Initial Access



3. Payload protection

1. CDR evasión via File Bloating for Initial Access



3. Payload protection

2. IPv4 Obfuscation and IT environments

- Easy to use
- Easy to implement and build your own script
- Really useful on IT environments.
- Easy to conceal and distribute in IT environments.
- Generally easy to work with in environments with strict DLP policies.
- Difficult to locate if you don't know the technique.

1 · SHELLCODE BYTES (X64 MSFVENOM CALC)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
FC	48	83	E4	F0	E8	C0	00	00	00	41	51	41	50	52	51

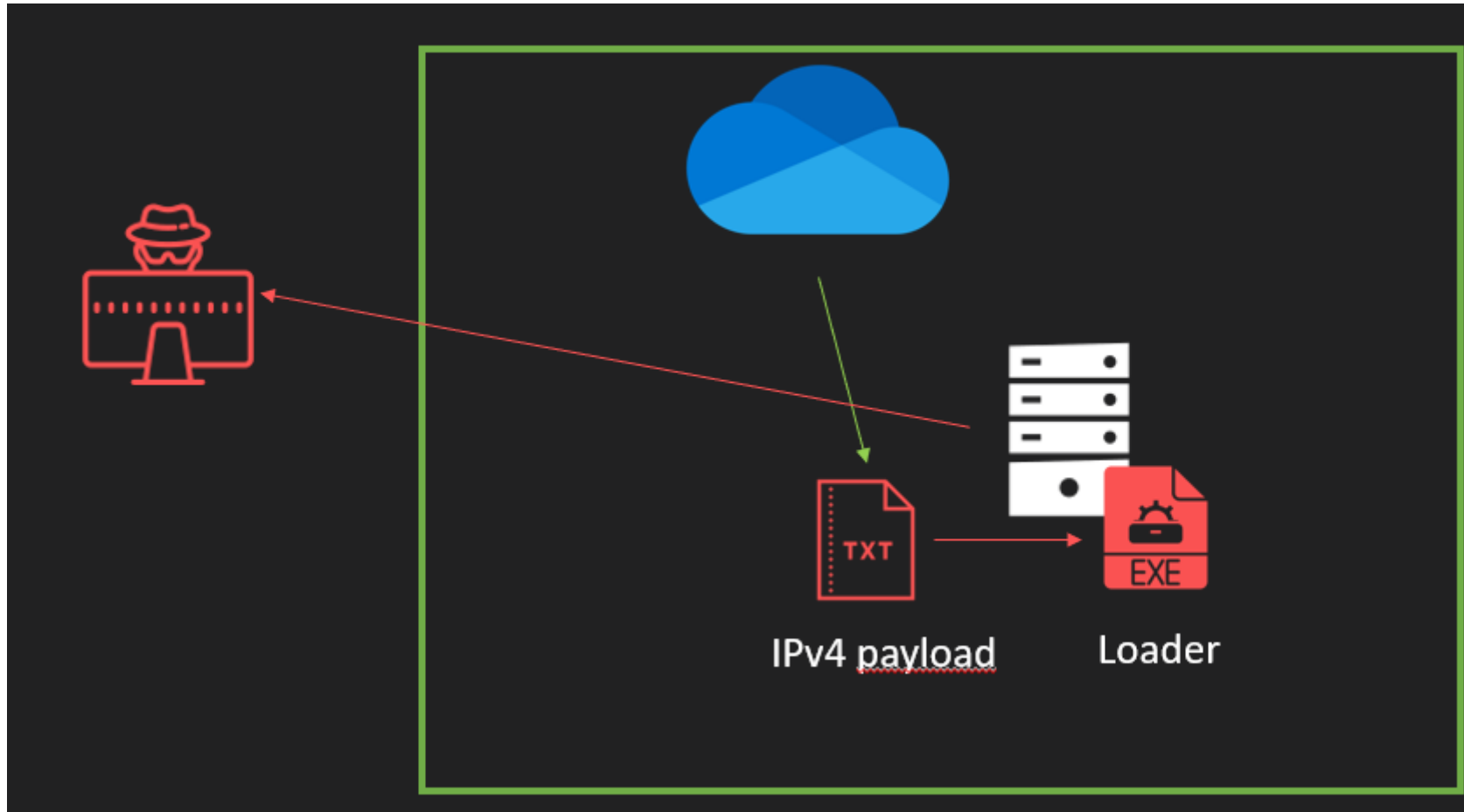
☒ Worked example (bytes 0-3) ☐ Following groups of 4 bytes

2 · EACH GROUP OF 4 BYTES → ONE IPV4 ADDRESS

FC hex → dec 252	48 hex → dec 72	83 hex → dec 131	E4 hex → dec 228	GROUP 1 ADDRESS = 252.72.131.228
F0 hex → dec 240	E8 hex → dec 232	C0 hex → dec 192	00 hex → dec 0	GROUP 2 ADDRESS = 240.232.192.0
00 hex → dec 0	00 hex → dec 0	41 hex → dec 65	51 hex → dec 81	GROUP 3 ADDRESS = 0.0.65.81
41 hex → dec 65	50 hex → dec 80	52 hex → dec 82	51 hex → dec 81	GROUP 4 ADDRESS = 65.80.82.81

3. Payload protection

2. IPv4 Obfuscation and IT environments



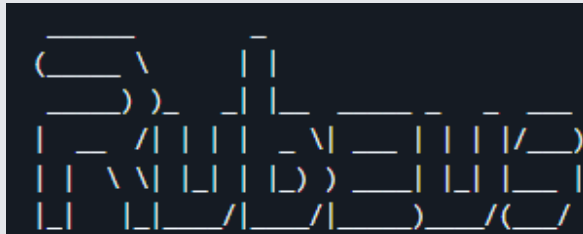
4. Tool protection

1. Behavior

2. Custom approaches

- **Mini-tooling**
- **Custom toolset**
- **Limit or delete Non-secure capabilities**

3. C# Tooling and Memory execution



[*] Action: Ask TGT

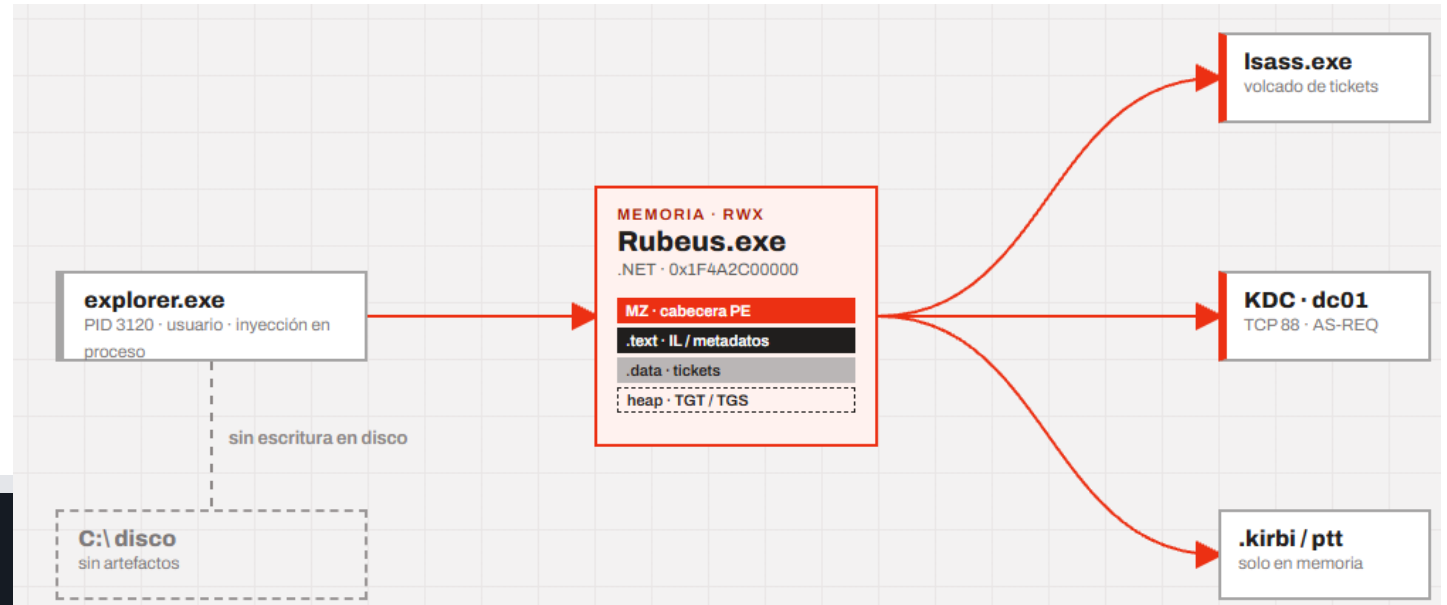
```
[*] Using rc4 hmac hash: 2b576acbe6bcfda7294d6bd18041b8fe
```

```
[*] Using domain controller: PRIMARY.testlab.local (192.168.52.100)
```

```
[*] Building AS-REQ (w/ preauth) for: 'testlab.local\dfm.a'
```

[+] TGT request successful!

```
[*] base64(ticket.kirbi):
```



4. Tool protection

HarmJ0y Added Rubeus.yar

protectionismachine Updating artifacts

Code Blame 14 lines (14 loc) · 722 Bytes

```
1 // From https://github.com/fireeye/red_team_tool_countermeasures/blob/3a773645093e77107df
2 // License: BSD 2-clause
3 rule HackTool_MSIL_Rubeus_1
4 {
5     meta:
6         description = "The TypeLibGUID present in a .NET binary maps directly to the Proj
7         md5 = "66e0681a500c726ed52e5ea9423d2654"
8         rev = 4
9         author = "FireEye"
10    strings:
11        $typelibguid = "658C8B7F-3664-4A95-9572-A3E5871DFC06" ascii nocase wide
12    condition:
13        uint16(0) == 0x5A4D and $typelibguid
14 }
```

Code Blame 28 lines (27 loc) · 1.36 KB

```
1 rule Windows_Hacktool_Rubeus_43f18623 {
2     meta:
3         author = "Elastic Security"
4         id = "43f18623-6024-4608-8019-e3fec54cf84"
5         fingerprint = "fbc2f67f394a4d21cac532b42c6749002cb7284b4a3912e18672881e6e74765d"
6         creation_date = "2022-10-20"
7         last_modified = "2022-11-24"
8         threat_name = "Windows.Hacktool.Rubeus"
9         reference_sample = "b7b4691ad1cdad7663c32d07e911a03d9cc8b104f724c2825fd4957007649235"
10        severity = 100
11        arch_context = "x86, arm64"
12        scan_context = "file, memory"
13        license = "Elastic License v2"
14        os = "windows"
15    strings:
16        $guid = "658C8B7F-3664-4A95-9572-A3E5871DFC06" ascii wide nocase
17        $print_str0 = "[*] Printing argument list for use with Rubeus" ascii wide
18        $print_str1 = "[+] Ticket successfully imported!" ascii wide
19        $print_str2 = "[+] Tickets successfully purged!" ascii wide
20        $print_str3 = "[*] Searching for accounts that support AES128_CTS_HMAC_SHA1_96/AES256_CTS_HMAC_SHA1_96" ascii wide
21        $print_str4 = "[*] Action: TGT Harvesting (with auto-renewal)" ascii wide
22        $print_str5 = "[X] Unable to retrieve TGT using tgtdeleg" ascii wide
23        $print_str6 = "[!] Unhandled Rubeus exception:" ascii wide
24        $print_str7 = "[*] Using a TGT /ticket to request service tickets" ascii wide
25    condition:
26        $guid or 4 of ($print_str*)
27 }
```

Yara Rules Analysis:

- Custom GUIDs
- References to the tool selected
- Output print

4. Tool protection

- Tool-name
- Project GUID
- PDB strings
- Name of some variables
- Method names as static indicator
- Strigs & Classes
- **Resume:** All static values and references to the tooling
- **More concise: Time** (And I don't have it)

```
[*] Threat found, splitting
[*] Testing 199403 bytes
[*] Threat found, splitting
[!] Identified end of bad bytes at offset 0x30A1B
000309EB 77 57 61 69 74 48 69 6E 74 00 64 77 43 68 65 63 wWaitHint.dwChec
000309FB 68 50 6F 69 6E 74 00 67 65 74 5F 43 6F 75 6E 74 kPoint.get_Count
00030A0B 00 4E 54 41 63 63 6F 75 6E 74 00 72 65 67 69 73 .NTAccount.regis
00030A1B 74 72 79 52 6F 6F 74 00 53 68 61 72 50 65 72 73 tryRoot.SharPers
00030A2B 69 73 74 00 52 65 67 69 73 74 72 79 50 65 72 73 ist.RegistryPers
00030A3B 69 73 74 00 49 57 73 68 53 68 6F 72 74 63 75 74 ist.IWshShortcut
00030A4B 00 4D 6F 76 65 4E 65 78 74 00 52 65 61 64 41 6C .MoveNext.ReadAl
00030A5B 6C 54 65 78 74 00 57 72 69 74 65 41 6C 6C 54 65 lText.WriteAllTe
00030A6B 78 74 00 67 65 74 5F 43 6F 6E 74 65 78 74 00 67 xt.get_Context.g
00030A7B 65 74 5F 4E 6F 77 00 67 65 74 5F 54 6F 64 61 79 et_Now.get_Today
00030A8B 00 73 65 74 5F 52 61 6E 64 6F 6D 44 65 6C 61 79 .set_RandomDelay
00030A9B 00 43 6F 6E 73 69 73 74 65 6E 63 79 00 50 72 6F .Consistency.Pro
00030AAB 63 65 73 73 65 64 42 79 46 6F 64 79 00 43 72 65 cessedByFody.Cre
00030ABB 61 74 65 53 75 62 48 65 79 00 4F 70 65 6E 53 75 ateSubKey.OpenSu
00030ACB 62 48 65 79 00 67 65 74 54 68 65 48 65 79 00 73 bKey.getTheKey.s
00030ADB 65 74 54 68 65 48 65 79 00 74 68 65 48 65 79 00 etTheKey.theKey.
[*] Run time: 15,32s
```

4. Tool protection

[h4wkst3r/InvisibilityCloak](#)

- Easily remove and obfuscate strings
- Change variables and information easily
- Fast!
- Easy to implement on CI/CD tool maintaining

```
C:\Toolkit\InvisibilityCloak-main>python InvisibilityCloak.py -d C:\source\KrbRelayUp-main -m reverse -n LetsGoUp

InvisibilityCloak

=====
[*] INFO: String obfuscation method: reverse
[*] INFO: Directory of C# project: C:\source\KrbRelayUp-main
[*] INFO: New tool name: LetsGoUp
=====

[*] INFO: Generating new GUID for C# project
[*] INFO: New project GUID is 4c3cf155-91d2-4c3e-a4d2-742b1b8bf336
[*] INFO: Changing C# project GUID in below files:

C:\source\KrbRelayUp-main\KrbRelayUp.sln
C:\source\KrbRelayUp-main\KrbRelayUp\KrbRelayUp.csproj

[*] INFO: Removing PDB string in C# project file

[*] INFO: Renaming KrbRelayUp.sln to LetsGoUp.sln
[*] INFO: Renaming KrbRelayUp.csproj to LetsGoUp.csproj
[*] INFO: Renaming directory KrbRelayUp to LetsGoUp

[+] SUCCESS: New GUID of 4c3cf155-91d2-4c3e-a4d2-742b1b8bf336 was generated and replaced in your project
[+] SUCCESS: New tool name of LetsGoUp was replaced in project

[*] INFO: Performing reverse obfuscation on strings in C:\source\KrbRelayUp-main\LetsGoUp\AskTGT.cs
[*] INFO: Performing reverse obfuscation on strings in C:\source\KrbRelayUp-main\LetsGoUp\KrbSCM.cs
[*] INFO: Performing reverse obfuscation on strings in C:\source\KrbRelayUp-main\LetsGoUp\Program.cs
[*] INFO: Performing reverse obfuscation on strings in C:\source\KrbRelayUp-main\LetsGoUp\S4U.cs
[*] INFO: Performing reverse obfuscation on strings in C:\source\KrbRelayUp-main\LetsGoUp\Asn1\Asn1Extensions.cs
[*] INFO: Performing reverse obfuscation on strings in C:\source\KrbRelayUp-main\LetsGoUp\Asn1\Asn1Elt.cs
[*] INFO: Performing reverse obfuscation on strings in C:\source\KrbRelayUp-main\LetsGoUp\Asn1\Asn1Exception.cs
[*] INFO: Performing reverse obfuscation on strings in C:\source\KrbRelayUp-main\LetsGoUp\Asn1\Asn1IO.cs
[*] INFO: Performing reverse obfuscation on strings in C:\source\KrbRelayUp-main\LetsGoUp\Asn1\Asn1OID.cs
[*] INFO: Performing reverse obfuscation on strings in C:\source\KrbRelayUp-main\LetsGoUp\Kerb\Crypto.cs
[*] INFO: Performing reverse obfuscation on strings in C:\source\KrbRelayUp-main\LetsGoUp\Kerb\Helpers.cs
```

4. Tool protection

[raskolnikov90/Beatrice.py](#)

- Generate patterns of simple assembly x64 instructions and their alternative instructions, turn them into machine code and patch the machine code if it matches.
- Build different lists of assembly instructions that contain immediate values and other instructions that can't be easily turned into patterns and apply appropriate changes to them.
- Apply alternative ways to encode instructions whenever possible.
- Create an identical binary functionality wise but with the above patches applied that will help evade YARA rules and some Antivirus solutions.
- Python → Easy to implement on CI/CD tool maintaining

```
C:\Users\Administrator\Documents>python beatrice.py demon.x64.exe

aaaaaaaa  aaaaaaaaa  aaaaaaa  aaaaaaaaa  aaaaaaaaa  aaaa  aaaaaaaaa  aaaaaaaaa  aaaaaaaaa  aaaa  aaaa
aaaaaaaa  aaaaaaaaa  aaaaaaaaa  aaaaaaa  aaaaaaaaa  aaaa  aaaaaaaaa  aaaaaaaaa  aaaaaaaaa  aaaa  aaaa
a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!  a!
!e  e!e  !e!  !e!  e!e  !e!  !e!  e!e  !e!  !e!  !e!  e!e  !e!  !e!  e!e  !e!  !e!  e!e  !e!  !e!  e!e  !e!
e!e!e!e  e!!!:  e!e!e!e!  e!!  e!e!e!e!  !!e  !e!  e!!!:  e!e!e!e!  !e!e!  !e!e!
!!!e!!!!  !!!!!:  !!!e!!!!  !!!  !!e!e!  !!!  !!!  !!!!!:  !!e!!!  e!!!
!!:  !!!  !!:  !!:  !!!  !!:  !!:  !!!  !!:  !!!  !!!  !!!  !!!  !!!  !!!  !!!  !!!  !!!  !!!  !!!  !!!
!:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:  !:
::  :::  ::  :::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::
::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::  ::

[+] Backup demon.x64.exe.bak created
[+] Detected PE file
[+] Generating patches
```

Thanks!

Contact & links:

cOnfig17.com

[Linkedin.com/in/antoniiodseri47](https://www.linkedin.com/in/antoniiodseri47)

github.com/cOnfig-17